

FAST-NUCES

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

CS-4112: Deep Learning Assignment 1

A Research Proposal on Diffusion Models for Synthetic Data Generation

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1. Problem Definition

1.1 Task Description

Diffusion models are a class of generative deep neural networks that learn to synthesize realistic data, primarily images, by progressively denoising random noise into structured outputs. The supervised learning framework in this context treats the denoising process as a prediction task: given a noisy image at time step t and the time step value itself, the deep model predicts the noise component added to the original clean image. This formulation makes diffusion-based image synthesis a supervised regression problem at its core.

The input to the model is a pair consisting of a partially noised image tensor and a scalar time step index. The output is a predicted noise map of identical spatial dimensions, which is then used to reconstruct a progressively cleaner version of the original image over hundreds of reverse diffusion steps. Optionally, in class-conditional or text-conditional settings, the model additionally receives a conditioning signal such as a class label or a text embedding and learns to generate images matching that condition.

According to Goodfellow et al. (2016), supervised learning involves learning a mapping from inputs to outputs based on example input-output pairs. The denoising objective in diffusion models precisely fits this definition: the model is trained with pairs of (noisy image, noise) where the noise map acts as the label, and the model minimizes a regression loss typically mean squared error between the predicted and actual noise. The learning paradigm is therefore supervised regression, embedded within an iterative generative framework.

1.2 Motivation and Relevance

The ability to generate high-fidelity, diverse synthetic images has enormous practical value. In healthcare, synthetic data generated by diffusion models can augment scarce medical imaging datasets, enabling training of diagnostic AI systems without violating patient privacy. In autonomous vehicle development, synthetic scenes can cover rare and dangerous scenarios that are difficult to capture in real data collection. In creative industries, diffusion models are already powering text-to-image generation tools used by millions of people worldwide.

Deep learning is particularly appropriate here for several reasons rooted in Goodfellow's Machine Learning Basics. First, raw images are extremely high-dimensional data: a 256×256 RGB image has over 196,000 dimensions making hand-crafted features fundamentally inadequate. Deep neural networks, through their layered non-linear transformations, are capable of learning compact representations that capture the underlying statistical structure of natural images. Second, the curse of dimensionality means that shallow or linear models cannot generalize from training examples to the full diversity of possible images. Deep architectures with hierarchical feature representations overcome this limitation by sharing statistical strength across spatial locations and semantic concepts.

Additionally, diffusion models have surpassed previous generative paradigms, including Generative Adversarial Networks (GANs), in terms of training stability, sample diversity, and distribution coverage. The field is rapidly evolving, with new architectural improvements and conditioning mechanisms being published continuously, making it a timely and impactful area for academic research.

1.3 Research Questions

This proposal is guided by the following core research questions:

- Can a deep neural network trained with a denoising diffusion objective produce synthetic images of sufficient quality to serve as viable training data for downstream classification tasks?
- How do recent architectural improvements in 2024–2025 diffusion models (such as flow matching, consistency distillation, and improved conditioning mechanisms) compare to their predecessors in terms of image quality metrics such as Fréchet Inception Distance (FID) and CLIP score?
- Can a simplified or distilled diffusion model achieve competitive image quality with significantly fewer sampling steps, making it practical for resource-constrained settings?

2. Literature Review

2.1 Overview

Deep learning has transformed the field of generative modeling over the past decade. Early approaches relied on Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs), which demonstrated that deep neural networks could learn to produce realistic images from random noise. However, these methods suffered from well known limitations: VAEs tend to produce blurry outputs due to their probabilistic latent space, while GANs are notoriously difficult to train due to mode collapse and adversarial instability.

Diffusion models emerged as a compelling alternative, introduced by Sohl-Dickstein et al. (2015) and refined by Ho et al. (2020) in the landmark Denoising Diffusion Probabilistic Models (DDPM) paper. The core idea progressively corrupting data with Gaussian noise and then learning to reverse this process provides a stable and principled training objective. As Goodfellow describes, the trend in deep learning has been toward methods that automatically learn representations from raw data rather than relying on hand engineered features; diffusion models embody this principle by learning the full data distribution through a learned denoising process.

Since 2020, the field has advanced rapidly. Latent Diffusion Models moved the diffusion process into a compressed latent space for efficiency. Score based models unified diffusion with stochastic differential equations. Conditional generation via CLIP embeddings and cross-attention mechanisms enabled text to image synthesis. The most recent advances in 2024–2025 focus on flow matching objectives, consistency models for faster sampling, and improved architectures such as DiT (Diffusion Transformer), which replaces the traditional U-Net backbone with transformer blocks.

2.2 Detailed Review of 3 Baseline Papers (2024+)

Baseline 1: Comprehensive Exploration of Diffusion Models in Image Generation: A Survey (2025)

Problem and Data:

This survey paper, published in Springer’s Machine Learning journal in 2025, provides a comprehensive examination of the diffusion model landscape, synthesizing findings across

hundreds of individual research papers. It does not evaluate a single dataset but instead aggregates results across standard benchmarks including ImageNet (1.28 million images, 1000 classes), COCO (330,000 images with captions), CIFAR-10, CelebA, and LSUN. The scope spans both unconditional and conditional image generation, covering text to image, class conditional, super-resolution, inpainting, and video generation tasks.

Deep Model Idea:

The survey categorizes diffusion models into several families: Denoising Diffusion Probabilistic Models (DDPMs), Score based Generative Models (SGMs), Latent Diffusion Models (LDMs), and Flow Matching models. Each employs a deep neural network typically a U-Net or Transformer architecture with several hidden layers that takes a noisy input and a time step embedding and predicts the noise or clean signal. The training objective is supervised regression with mean squared error as the reconstruction loss.

Results and Metrics:

The survey reports that state of the art diffusion models achieve FID scores below 2.0 on ImageNet 256×256 class conditional generation, with the best models reaching FID 1.5. On text to image benchmarks using CLIP score and human preference studies, models like DALL-E 3 and Stable Diffusion XL achieve strong alignment between generated images and text prompts. Sampling speed remains a limitation: standard DDPM requires 1000 denoising steps, but DDIM and consistency models reduce this to 10–50 steps with minimal quality loss.

Strengths:

The survey’s breadth makes it an invaluable reference for understanding the entire field at once. It clearly organizes the rapidly growing literature and provides comparative tables of FID scores across methods and datasets, making it easy to identify the best performing approaches.

Weaknesses and Gaps:

As a survey, it does not introduce a new method or provide experimental reproducibility details. Some of the referenced methods are evaluated on proprietary or closed datasets, making direct comparison difficult. The survey also covers a very wide range of tasks without deeply analyzing any single application domain, leaving room for focused experimental work.

Baseline 2: SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis (2024)

Problem and Data:

This paper addresses the problem of high-resolution text to image synthesis, focusing on producing photorealistic 1024×1024 images from natural language descriptions. The model is trained on a large curated dataset of image-text pairs, primarily from the LAION-5B dataset (approximately 5 billion image-text pairs), filtered for aesthetic quality and safety. Evaluation is conducted on the COCO validation set and through human preference studies comparing generated images against reference outputs.

Deep Model Idea:

SDXL extends the Latent Diffusion Model framework by using a significantly larger U-Net backbone with several additional cross-attention layers to incorporate text conditioning. The model operates in the latent space of a pretrained variational autoencoder, compressing images into a 128×128 latent representation before applying the diffusion process. A novel refinement stage uses a second deep neural network to further improve image details by applying diffusion in pixel space on high-frequency features.

Results and Metrics:

SDXL achieves state of the art performance at the time of publication, with significantly

higher human preference ratings compared to Stable Diffusion 1.5 and 2.1. On COCO FID benchmarks, it achieves FID scores competitive with or superior to DALL-E 2 while being open source. The CLIP score on MS-COCO captions improves substantially over prior versions of Stable Diffusion.

Strengths:

SDXL is open source, making it highly reproducible. Its two-stage architecture (base + refiner) and the use of multiple conditioning inputs (text, image size, crop coordinates) represent significant engineering contributions that improve output quality without fundamentally changing the training paradigm.

Weaknesses and Gaps:

The model is computationally expensive, requiring multiple high-end GPUs for training and significant VRAM for inference. The training data composition and filtering criteria are not fully disclosed, making it difficult to assess potential biases. Additionally, the model still struggles with accurate text rendering within generated images and complex spatial reasoning tasks.

Baseline 3: Consistency Models (Song et al., 2024 Extended Version)

Problem and Data:

This work addresses the fundamental sampling inefficiency of diffusion models, which require hundreds of sequential neural network evaluations to generate a single image. The goal is to train a deep model that can generate high-quality images in as few as one or two network evaluations. Evaluation is performed on CIFAR-10 (50,000 training images, 32×32 resolution) and ImageNet 64×64 (1.28 million images), which are standard benchmarks for generative model comparison.

Deep Model Idea:

Consistency models train a deep neural network to predict the clean endpoint of the diffusion trajectory directly from any intermediate noisy point, enforcing a self-consistency constraint that outputs at different noise levels should map to the same clean image. This is achieved through either distillation from a pretrained diffusion model or direct consistency training, both using a specialized loss function derived from the diffusion probability flow ODE.

Results and Metrics:

Consistency distillation achieves FID scores of 3.55 on CIFAR-10 with a single sampling step, compared to DDPM’s 3.17 with 1000 steps a remarkable result showing that a single forward pass can nearly match thousands of steps. With two sampling steps, FID improves to 2.93, surpassing DDPM. On ImageNet 64×64 , consistency models achieve competitive FID scores while reducing sampling time by $50\times$.

Strengths:

The consistency model framework is elegant, theoretically grounded, and practically impactful reducing inference time from minutes to seconds. It is compatible with existing diffusion model architectures, making it easy to apply to pretrained models.

Weaknesses and Gaps:

The quality gap between consistency models and full-step diffusion models widens at higher resolutions. Direct consistency training (without distillation) is more difficult to stabilize and tends to produce slightly lower quality outputs than distillation based approaches. The method has primarily been validated on standard benchmarks and may not transfer uniformly to specialized domains such as medical imaging.

2.3 Short Summaries of 7 Additional Papers

Paper 4: Denoising Diffusion Probabilistic Models — Ho et al. (2020)

This foundational paper introduced the DDPM framework that underpins most modern diffusion models. It formulates image generation as a Markov chain of Gaussian noise additions followed by learned denoising steps, with the deep model trained on a simplified noise-prediction objective. The paper demonstrated FID of 3.17 on CIFAR-10 and established the evaluation protocol used by nearly all subsequent work. It remains essential for understanding the mathematical foundation of the denoising objective used in this proposal.

Paper 5: Improved Denoising Diffusion Probabilistic Models — Nichol & Dhariwal (2021)

This work improved upon DDPM by learning the variance of the reverse diffusion process (not just the mean), enabling faster sampling with fewer steps. It also introduced classifier-free guidance, a technique that substantially improves conditional generation quality by training a single deep network to jointly model conditional and unconditional distributions. Classifier-free guidance is now universally adopted in text to image models including SDXL and Stable Diffusion, making this paper essential background for understanding modern conditional generation.

Paper 6: High-Resolution Image Synthesis with Latent Diffusion Models — Rombach et al. (2022)

This paper introduced Latent Diffusion Models (LDMs), which apply the diffusion process in the compressed latent space of a pretrained autoencoder rather than directly on pixel values. This reduces computational cost dramatically, enabling high-resolution image generation on consumer hardware. LDMs form the backbone of Stable Diffusion, one of the most widely deployed open source generative models. Understanding LDMs is essential for this proposal, as SDXL (Baseline 2) is a direct extension of this work.

Paper 7: Scalable Diffusion Models with Transformers (DiT) — Peebles & Xie (2023)

DiT replaces the traditional U-Net backbone in diffusion models with a pure Transformer architecture applied to patches of the latent image. The paper demonstrates that Transformer based diffusion models scale more efficiently than U-Net based ones: larger DiT models achieve better FID scores on ImageNet with predictable performance improvements as model size increases. DiT-XL/2 achieves FID of 2.27 on ImageNet 256×256 , setting a new benchmark at the time. This architecture is relevant because several 2024 state of the art models build directly on DiT.

Paper 8: DALL-E 3: Improving Image Generation with Better Captions — Betker et al. (2023)

This paper focuses on improving text-image alignment in large-scale text to image diffusion models by recaptioning training data with detailed, accurate descriptions using a large language model. The finding that data quality and caption accuracy matter more than model scale alone is highly relevant for understanding how to improve conditional generation. DALL-E 3 achieves state of the art CLIP scores on prompt-following benchmarks while maintaining strong FID performance. This motivates careful attention to conditioning quality in proposed approaches.

Paper 9: Score-Based Generative Modeling Through Stochastic Differential Equations — Song et al. (2021)

This paper provides a unified mathematical framework connecting DDPM, NCSN, and other diffusion approaches through the lens of stochastic differential equations (SDEs). It shows that all diffusion models can be understood as discretizations of a continuous SDE, and that the reverse process is also a diffusion SDE with a learned score function. This theoretical unification motivated subsequent work on flow matching and consistency models, and is important for understanding the mathematical foundations cited in the survey (Baseline 1).

Paper 10: Elucidating the Design Space of Diffusion-Based Generative Models — Karras et al. (2022)

This paper systematically analyzes the design choices in diffusion model training and inference, including noise schedules, network preconditioning, and sampling algorithms. It introduces the EDM (Elucidated Diffusion Model) framework, achieving FID of 1.79 on CIFAR-10 the best result at the time through careful principled choices rather than architectural novelty. The paper provides actionable guidance for practitioners designing diffusion model training pipelines, directly informing the evaluation and experimental setup described in this proposal.

3. Proposed Deep Learning Approach

3.1 High-Level Idea

We propose to train a deep neural network on a standard image benchmark dataset using the denoising diffusion probabilistic model objective, augmented with class conditioning to enable controlled image generation. The core idea is straightforward: during training, the model receives noisy images (corrupted by a known amount of Gaussian noise) together with their corresponding class labels and learns to predict the added noise. During inference, the model starts from pure random noise and iteratively denoises it over many steps, guided by the class conditioning signal, until a realistic image emerges.

Our proposal extends the baseline DDPM framework with two key modifications inspired by the reviewed literature: (1) we incorporate classifier-free guidance (following Nichol & Dhariwal, 2021) to improve class-conditional generation quality, and (2) we explore consistency distillation (following Song et al., 2024) to reduce the number of required sampling steps from 1000 to 10–50, dramatically reducing inference time. This combination allows us to compare our approach against all three baselines on shared metrics.

3.2 Input Representation

The input data consists of images normalized to have pixel values in the range $[-1, 1]$. For class-conditional generation, each image is paired with an integer class label that is embedded into a learned vector representation through a trainable embedding layer. Time step indices are similarly embedded using sinusoidal positional encodings combined with learned transformations, following the approach established in DDPM and used in all subsequent diffusion models. If we operate in latent space (following the LDM approach), images are first encoded into a compressed latent representation using a pretrained variational autoencoder, reducing the spatial dimensions by a factor of 8 in each dimension.

3.3 Network Outline

Our proposed deep neural network follows an encoder-decoder structure with skip connections, containing approximately 12–18 hidden layers organized in a hierarchical multi-scale design. The network takes as input the concatenation of the noisy image features and the time step embedding, processes them through several downsampling blocks that progressively reduce spatial dimensions while increasing feature depth, and then reconstructs the noise map through symmetrical upsampling blocks. Class conditioning is incorporated via adaptive layer normalization, which modulates the feature maps at each hidden layer based on the class embedding.

Non-linearity is introduced throughout using the SiLU (Sigmoid Linear Unit) activation function, which provides smooth gradients beneficial for the regression objective. The network is trained end-to-end using the Adam optimizer with a cosine learning rate schedule. The training loss is mean squared error (MSE) between the predicted and actual noise maps, which corresponds to the simplified DDPM training objective. Regularization is applied through dropout and exponential moving average (EMA) of model weights, following standard practice in diffusion model training.

3.4 Why This Approach is Suitable

As Goodfellow et al. emphasize, deep neural networks excel at learning hierarchical representations of complex high-dimensional data. Images belong to a manifold of much lower intrinsic dimensionality than their raw pixel dimension, and deep networks with many layers of non-linear transformations are uniquely capable of discovering and exploiting this structure. The denoising objective is a particularly effective way to train such networks because it provides dense, informative supervision at every training step — the model must simultaneously understand local textures, global structures, and semantic content in order to accurately predict the noise component.

The incorporation of consistency distillation addresses the primary practical limitation of diffusion models (slow sampling) while maintaining the high quality achieved through the full diffusion training. This positions our approach at the intersection of quality and efficiency, directly addressing gaps identified in all three baseline papers.

4. Dataset Description

4.1 Source of Data

We will use two complementary public datasets for our experiments, selected to allow comparison with both the baselines and the broader literature:

- **CIFAR-10** (<https://www.cs.toronto.edu/~kriz/cifar.html>): A widely used benchmark consisting of 60,000 color images across 10 object classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck), split into 50,000 training and 10,000 test images, each at 32×32 pixel resolution.
- **ImageNet 64×64** (<https://image-net.org>): A downsampled version of the full ImageNet dataset containing 1.28 million training images and 50,000 validation images across 1000 classes, resized to 64×64 pixels. This allows comparison with Consistency Models (Baseline 3) and provides a more challenging evaluation environment.

4.2 Size and Structure

CIFAR-10 provides 50,000 training samples and 10,000 test samples. Each sample is a $32 \times 32 \times 3$ array of pixel intensity values (RGB channels), paired with an integer class label from 0 to 9. The dataset is perfectly balanced with 5,000 examples per class in the training set. ImageNet 64×64 provides 1.28 million training samples, each a $64 \times 64 \times 3$ image paired with one of 1000 class labels. Both datasets are supervised in the sense that class labels are available and are used for conditional generation training.

4.3 Preprocessing

All images are normalized from integer pixel values in $[0, 255]$ to floating-point values in $[-1, 1]$ by applying the transformation:

$$x_{normalized} = \left(\frac{x}{127.5} \right) - 1.0$$

This normalization is standard for diffusion model training as it centers the data around zero and gives unit-scale inputs. Random horizontal flipping is applied as a data augmentation during training to improve generalization. No complex preprocessing beyond this is required, as diffusion models operate directly on normalized image tensors. Class labels are represented as integer indices and embedded into 512-dimensional vectors via a learned embedding table.

4.4 Why These Datasets Fit the Problem

CIFAR-10 is the canonical benchmark for evaluating generative models at small scale and is used by DDPM (Paper 4), Improved DDPM (Paper 5), EDM (Paper 10), and Consistency Models (Baseline 3), enabling direct comparison of FID scores. Its small image size and manageable training set size make it ideal for rapid experimentation and hyperparameter tuning within the computational constraints of an academic setting. ImageNet 64×64 provides a more realistic and challenging benchmark for evaluating the scalability of our approach, and is used in the Consistency Models baseline, ensuring direct comparability.

5. Evaluation Plan

5.1 Metrics

We will evaluate generated image quality using the following metrics, chosen to match those used by the baseline papers:

- **Fréchet Inception Distance (FID):** The primary metric for evaluating generative model quality. FID measures the statistical distance between the distribution of generated images and real images in the feature space of a pretrained Inception-v3 network. Lower FID indicates higher quality and diversity. FID is used by all three baseline papers and virtually all published diffusion model work.
- **Inception Score (IS):** Measures both the quality (sharpness) and diversity of generated images by evaluating the predictability and entropy of class predictions from a pretrained classifier. Higher IS indicates better quality. Used alongside FID in DDPM and Improved DDPM.

- **Precision and Recall for Generative Models:** Precision measures the fraction of generated samples that fall within the real data manifold (fidelity), while Recall measures the fraction of the real data manifold covered by generated samples (diversity). This decomposition is important for understanding whether quality improvements come at the cost of mode coverage.
- **Sampling Speed (NFE):** Number of function evaluations (network forward passes) required to generate a single image. Lower NFE indicates faster sampling. This is particularly important for comparing our consistency-distilled model against the standard DDPM baseline.

5.2 Experimental Setup

Data splits:

For CIFAR-10, we use the standard 50,000/10,000 training/test split. FID is computed using 50,000 generated samples compared against the full training set, following standard practice. For ImageNet 64×64 , FID is computed using 50,000 generated samples compared against the validation set. We additionally hold out 10% of the CIFAR-10 training set as a validation set for hyperparameter tuning and early stopping.

Training plan:

Models will be trained for up to 800,000 gradient steps on CIFAR-10 and up to 1.5 million steps on ImageNet 64×64 , following the training schedules from DDPM and Consistency Models. We will use early stopping based on validation FID, computed every 50,000 steps. Batch size will be set to 128 for CIFAR-10 and 256 for ImageNet 64×64 , which fits within the memory constraints of standard research GPU configurations (e.g., NVIDIA A100 40GB).

Hyperparameters to tune:

- Learning rate: $\{1e-4, 2e-4, 5e-4\}$ — tested with linear warmup and cosine decay
- Number of diffusion steps: $\{500, 1000\}$ during training; $\{10, 20, 50, 1000\}$ during inference
- Classifier-free guidance scale: $\{1.0, 1.5, 2.0, 3.0\}$
- Network depth: $\{12, 15, 18 \text{ hidden layers}\}$ — testing the effect of depth on performance
- EMA decay rate: $\{0.9999, 0.9995\}$

5.3 Baseline Comparison

We will compare our proposed model against the three baseline methods on shared benchmarks using identical evaluation protocols:

- **vs. Baseline 1 (Survey):** We will position our FID and IS scores on CIFAR-10 and ImageNet 64×64 within the landscape reported in the survey, identifying which family of methods our approach most closely resembles and where it improves or falls short.
- **vs. Baseline 2 (SDXL):** While SDXL operates at much higher resolution and scale, we will replicate its core architectural innovations (cross-attention conditioning, two-stage generation) at CIFAR-10 scale and compare FID and human preference ratings on a small-scale text-conditional variant.

- **vs. Baseline 3 (Consistency Models):** We will directly compare FID vs. NFE tradeoff curves on CIFAR-10 and ImageNet 64×64 , testing our consistency-distilled variant against the published consistency model results under identical evaluation protocols.

6. Expected Outcomes

6.1 Performance Expectations

We expect our baseline DDPM implementation to reproduce the published FID scores from Ho et al. (2020) within a small margin (0.3 FID points), validating our experimental pipeline. With the addition of classifier-free guidance (Nichol & Dhariwal, 2021), we expect to achieve FID below 3.5 on CIFAR-10 at 1000 sampling steps, matching or improving upon Improved DDPM’s reported results.

For our consistency-distilled variant, we expect to achieve FID below 5.0 on CIFAR-10 with only 10 sampling steps — a significant quality-efficiency tradeoff improvement over standard DDPM that requires 1000 steps for comparable quality. On ImageNet 64×64 , we expect FID below 6.0 with 10 steps, competitive with the published Consistency Models results.

6.2 Learning and Understanding Outcomes

Through this project, we expect to develop practical expertise in the following deep learning concepts, building on the theoretical foundations of Goodfellow’s textbook:

- **Effect of depth on generative model quality:** We will observe how adding more hidden layers improves the model’s ability to capture complex image statistics, consistent with Goodfellow’s discussion of depth and representation power.
- **Learned representations:** The intermediate activations of our trained model will encode semantic structure — the same deep network that predicts noise at early (high-noise) time steps must understand high-level semantics, while at late (low-noise) steps it captures fine details. This demonstrates hierarchical representation learning in practice.
- **Sensitivity to hyperparameters:** We will empirically characterize how learning rate, guidance scale, and number of sampling steps affect output quality, providing intuition for practical diffusion model deployment.
- **Training dynamics of deep generative models:** We will observe the relationship between training loss and FID, noting that loss alone is insufficient to characterize generative quality — a key insight distinguishing generative modeling from discriminative tasks.

6.3 Potential Limitations

Several limitations must be acknowledged in advance:

- **Computational constraints:** Training state-of-the-art diffusion models at full scale requires thousands of GPU-hours. Our experiments will be limited to CIFAR-10 and ImageNet 64×64 , and we cannot replicate the full SDXL training at 1024×1024 resolution. Results at lower resolution may not fully generalize to high-resolution settings.

- **Limited dataset diversity:** CIFAR-10 and ImageNet 64×64 are standard benchmarks but are less complex than real-world deployment scenarios. The synthetic data quality may degrade in specialized domains with limited training data (e.g., medical imaging, satellite imagery).
- **Simplifications vs. original papers:** We will approximate the SDXL baseline at a smaller scale, which may miss emergent behaviors that only appear at the scale of billions of training samples. Our comparisons with SDXL will therefore be indicative rather than definitive.
- **Evaluation metric limitations:** FID is known to have limitations as a perceptual quality metric — it can be low for models that memorize training data, and does not fully capture human aesthetic preferences. We will supplement FID with IS and Precision/Recall to provide a more complete picture.

References

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- [4] Peebles, W., & Xie, S. (2023). Scalable diffusion models with transformers. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 4195–4205.
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- [9] Song, Y., Dhariwal, P., Chen, M., & Sutskever, I. (2024). Consistency models (extended version). *arXiv:2303.01469v3*.

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- [11] Podell, D., English, Z., Lacey, K., Blattmann, A., Dockhorn, T., Müller, J., ... & Rombach, R. (2024). SDXL: Improving latent diffusion models for high-resolution image synthesis. *International Conference on Learning Representations (ICLR)*.

Paper List Summary

The following 10 papers are included in this proposal’s literature review:

Baseline Papers (2024+):

- [10] **Baseline 1:** Yang et al. (2025) — Comprehensive Exploration of Diffusion Models in Image Generation: A Survey. *Springer Machine Learning*. DOI: 10.1007/s10462-025-11110-3
- [SDXL] **Baseline 2:** Podell et al. (2024) — SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis. *ICLR 2024*.
- [9] **Baseline 3:** Song et al. (2024) — Consistency Models (Extended Version). *arXiv:2303.01469v3*.

Additional Related Papers (7):

- [1] Ho et al. (2020) — Denoising Diffusion Probabilistic Models. *NeurIPS 2020*.
- [2] Nichol & Dhariwal (2021) — Improved Denoising Diffusion Probabilistic Models. *ICML 2021*.
- [3] Rombach et al. (2022) — High-Resolution Image Synthesis with Latent Diffusion Models. *CVPR 2022*.
- [4] Peebles & Xie (2023) — Scalable Diffusion Models with Transformers (DiT). *ICCV 2023*.
- [5] Betker et al. (2023) — Improving Image Generation with Better Captions (DALL-E 3). *OpenAI 2023*.
- [6] Song et al. (2021) — Score-Based Generative Modeling Through Stochastic Differential Equations. *ICLR 2021*.
- [7] Karras et al. (2022) — Elucidating the Design Space of Diffusion-Based Generative Models (EDM). *NeurIPS 2022*.